

Grégoire Pelletier

Current research: Sparse Autoencoders for interpretability of text embeddings (EDF R&D)

Machine Learning Researcher | Mechanistic Interpretability | Representation Learning

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RESEARCH INTERESTS

- Mechanistic Interpretability of Large Language Models (SAE, superposition, circuits)
- Representation Learning and Feature Geometry in Neural Networks
- Sparse Coding and Monosemantic Representations
- AI Safety via Interpretability and Robustness

RESEARCH & PROFESSIONAL EXPERIENCE

EDF R&D (SEQUOIA - E7S) – *Machine Learning Research Intern*
Saclay, France

Mar 2026 – now

- Study and implementation of Sparse Autoencoders (SAE) for interpreting dense text embeddings on real industrial datasets
- Investigation of latent feature structure and sparsity-induced disentanglement in language representations
- Exploration of feature alignment with semantic concepts and potential causal structure
- Design of data-efficient interpretability pipelines under limited domain-specific data constraints

Paris Digital Lab – *Machine Learning Engineer*
La Défense, France

Feb 2025 – Jul 2025

- Designed LLM-based systems interfacing with structured APIs (SNCF)
- Developed multimodal retrieval system (RAG) with visual document understanding
- Built end-to-end pipelines for LLM-based knowledge extraction and generation

Emotors (Stellantis) – *Machine Learning Intern*
Carrières-sous-Poissy, France

Jun 2024 – Nov 2024

- Designed a thermal estimator for electric machines using recurrent neural networks (RNN, PyTorch)
- Adapted Neural Network-based physical modeling approaches (Thermal Neural Networks)

ENS Paris-Saclay (Centre Borelli) – *Statistical Research Intern*
Gif-sur-Yvette, France

May 2023 – Jul 2023

- Modeled textures using Gaussian fields and applied neural networks to image-based statistical data

Laboratoire de Mathématiques d'Orsay – *Supervised Research*
Orsay, France

Feb 2023 – Jun 2023

- Studied random graphs and community detection using the Hashimoto matrix

EDUCATION

Université Paris-Saclay

Sep 2022 – now

Magistère of Mathematics & Master's Degree in Mathematics and Artificial Intelligence

Relevant coursework: Probability, Measure Theory, Optimization, Deep Learning, Reinforcement Learning, Statistical Learning, Online Learning and Game Theory, Sequential Learning, Supervised Learning, Unsupervised Learning.

Lycée Louis-le-Grand

Sep 2019 – Jun 2022

Classe Préparatoire aux Grandes Écoles (Mathematics and Physics)

PROJECTS & INDEPENDENT STUDY

Research on Mechanistic Interpretability of Sparse Representations

- Conducted an in-depth study of Sparse Autoencoders and superposition in language models
- Analyzed recent architectures (JumpReLU, Matryoshka SAE) and evaluation methods for monosemanticity
- Proposed a dual micro/macro SAE framework for interpretable representation learning under data constraints

SKILLS & LANGUAGES

Programming:

Python, R, OCaml, SQL, LaTeX, Spark

Machine Learning:

PyTorch, TensorFlow, Scikit-learn, Pandas

Languages:

French (Native), English (C1 - Linguaskill), German